



Full length article

Dynamic thermal evaluation of a green roof system under warm weather conditions: Case of Tropical and Dry climates

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ABSTRACT

This study presents the thermal evaluation of a green roof (GR) under the climatic conditions of two typical warm climates in Mexico: dry arid (*BWh*) and humid tropical (*Af*). The thermal behavior of the analyzed system was assessed using a numerical code based on the finite volume method, which was effectively validated with experimental data, showing less than 6% difference in volumetric water content (VWC) and surface temperature on the substrate. The GR was evaluated over the warmest and coldest periods of the year; its performance was compared to that of a conventional concrete roof (RR). The results indicated that the GR system significantly reduces energy gains by up to 90 and 94% during the warmest period in the *BWh* and *Af* climates, respectively. For the *Af* climate, the VWC decreased by 18% during the warm period and by 4% in the cold period. For the *BWh* climate, there was a reduction of 17.1% in the cold period and 60% in the warm period. This finding highlights the need for substrate irrigation to maintain evaporation energy dissipation in the arid climate during the summer.

1. Introduction

According with Nejat et al., at present, buildings consume 40% of global energy [1]. Furthermore, this sector has increased 14% its consumption of energy over the last decade [2]. This leads to buildings responsible for 33% of greenhouse gas emissions, which contributes to global warming. Both scientific communities and companies have identified the need to reduce energy demand in buildings through novel techniques applied to the envelope to reduce energy consumption in this sector [3,4]. In particular, the roof of a building is the component that is most exposed to solar radiation. Thus, it is necessary to apply technologies in this component that help reduce energy consumption and mitigate the urban heat island phenomena (UHI). There are various roofing technologies that reduce energy gain to a building. Among the most popular are cool roofs (CR), roofs with radiative cooling materials (RCR), roofs with phase change materials (PCMR), and green roofs (GR). Each of them allows to reduce the energy gain based on specific characteristics: CRs are characterized by being highly reflective of solar radiation, RCRs reflect and emit thermal radiation preferably in the infrared, to maximize heat loss to the celestial vault, PCMR's reduce the fluctuation of the roof temperature considerably, and finally, GRs reduce the energy gain through a shading effect as well as thermal insulation [5–9]. Each of these systems has different advantages and limitations. Table 1 shows a comparison of the main characteristics of these technologies. It is observed

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Nomenclature

a	drag coefficient on the substrate surface (m)
b	hydraulic parameter (-)
c_p	specific heat (J/kgK)
d_0	displacement height (m)
D_θ	hydraulic diffusivity (m ² /s)
d_f	mean leaf size (m)
G_{sol}	solar radiation on the system (W/m ²)
h	convective coefficient (W/m ² K)
H_f	vertical distance between the soil and the foliage (m)
H_g	thickness of the soil (m)
H_{slab}	thickness of the concrete slab (m)
j_{dr}	mass flux due to drain (kg/m ² s)
j_{eva}	mass flux due to evotranspiration (kg/m ² s)
j_{pr}	mass flux due to irrigation (kg/m ² s)
k	Von Karman constant (-)
k_θ	hydraulic conductivity (m/s)
k_e	Kersten number (-)
l_{fg}	latent heat of vaporization (J/kg)
LAI	leaf area index (m ² /m ²)
N_z	number of computational nodes (-)
P_a	air vapor pressure (Pa)
$P_{f,sat}$	foliage saturation pressure (Pa)
$P_{g,sat}$	substrate saturation pressure (Pa)
q	heat flux (W/m ²)
Q_{gain}	net heat gain (J/m ²)
R	net radiative heat flux (W/m ²)
r_a	aerodynamic resistance to heat transfer (m/s)
r_c	resistance to heat transfer between the substrate and the foliage (m/s)
r_{stoma}	stomatal resistance (m/s)
r_{vap}	substrate surface resistance to mass transfer (m/s)
T	temperature (K)
T_r	temperature of the conventional roof (K)
T_{sky}	sky temperature (K)
u	wind speed (m/s)
u_f	wind velocity over foliage surface (m/s)
u_g	wind velocity over substrate surface (m/s)
z	vertical position (m)
Z_0	roughness (m)
Z_M	roughness lenght for momentum (m)
Z_u	stability correction at the top of the canopy (m)

Greek

α^*	absorptivity (-)
Δt	time step (s)
ϵ^*	emissivity (-)
γ	thermodynamic psychrometric constant (Pa/K)
λ	thermal conductivity (W/mK)
ψ_{sat}	moisture potential of soil tension when soil is saturated (m)
ρ	density (kg/m ³)
ρ^*	reflectivity (-)
σ	Stefan–Boltzman’s constant (W/m ² K ⁴)

τ_f^*	leaf shortwave transmissivity (-)
θ	volumetric water content (kg/m ³ /kg/m ³)
Subscripts	
<i>a</i>	ambient air
<i>cond</i>	conduction
<i>conv</i>	convection
<i>dry</i>	dry condition
<i>eva</i>	evaporative
<i>f</i>	foliage
<i>g</i>	substrate
<i>in</i>	interior
<i>rad</i>	radiation
<i>s</i>	slab
<i>sat</i>	saturation condition
Abbreviations	
GR	Green roof
RR	Reference roof

Table 1
Characteristics of technologies used for roofing.

Feature	Green roofs	Cool roofs	Radiative cooling materials	Phase change materials
Primary function	Insulation, cooling shading	Reflective cooling	Reflective and radiative cooling	Thermal energy storage
Energy saving	Seasonal, moderate to high	High in summer	High in summer	Year-round, moderate
UHI mitigation	High (reduces air temperature)	Moderate (localized cooling)	Moderate (localized cooling)	Low (indirect impact)
Seasonal performance	Year-round	Summer only	Summer only	Year-round
Environmental impact	Positive (biodiversity, air quality)	Neutral	Neutral	Neutral
Cost	High upfront, moderate upkeep	Low upfront, low maintenance	Moderate upfront, low maintenance	Moderate upfront, low maintenance
Additional benefits	Stormwater management, aesthetics	Simple and durable	Low maintenance	Enhanced indoor comfort

that GR systems offer a comprehensive solution that balances energy savings with ecological and social benefits. Therefore, they are an excellent option compared to other technologies when project priorities include improving urban ecosystems, reducing the UHI effect, and creating functional and visually attractive spaces.

A green roof is a partially or totally covered roof with vegetation that provides aesthetic, economic, and environmental benefits. One of the main advantages of GR in reducing energy consumption used to achieve comfort is a characteristic that no other options have: Evapotranspiration mechanisms contribute significantly to heat loss to the environment by up to 58.4% [10]. This peculiarity helps to mitigate the effect of the UHI. Nowadays, the GR studies have increased due to the multiple benefits they provide [11]. However, the GR benefits hinge on the climate, geometry, materials composition, and volumetric water content (VWC). Due to the multiple factors that impair the performance of GR, it is necessary to investigate their throughput in different climates and settings to obtain the most significant benefits or determine their limitations. Thus, since climate and VWC are two of the main factors that influence the operation of the GR in terms of energy savings and functionality, it becomes relevant to study the effect that these two variables have on the heat transfer in the building envelope [12,13].

Next, studies related to the thermal behavior of GR in different climates are commented on, which were assorted according to the Köppen–Geiger climate classification presented by Beck et al. [14]: temperate (C), tropical (A), dry (B) and continental (D) climates.

There are numerous studies on implementing green roof systems in different climates. In the case of temperate climates (C), it was found that in a city with a Cwa climate, GR increases the insulation effect during the winter and reduces the entry of energy into the interior during the

summer [15]. This behavior was also observed by Bevilacqua et al. in a Csa climate; during the summer, the substrate had an average temperature of 12 °C lower than that of a typical roof, while in winter it remained 4 °C higher [16]. Additionally, the use of GR impacts on the reduction of energy consumption by up to 76% [17]. On the other hand, the implementation of GR in tropical climates (A) has allowed us to observe that GR functions as an energy sink most of the day during warm periods, while, in cold periods, GR increases thermal mass and reduces temperature fluctuation inside [18,19]. The GR system has also been studied under continental-type climates (D), and it has been reported that even in winter periods when snow covers the roof, the thermal behavior of a GR is remarkably like that of a conventional roof [20,21]. Tang et al. theoretically and experimentally studied a GR system under Dfa climatic conditions during winter. The authors found that water within the substrate can change phase and thus influence the system's thermal performance [22]. Finally, the use of GR systems in arid climates (B) reduces indoor air temperature by up to 13 °C compared to a conventional roof [23,24] and considerably reduces the temperature oscillation of the concrete [25].

Chagolla et al. reported one of the first works to analyze irrigation's effect on a GR system's thermal performance [26]. This study was conducted under subtropical climatic conditions and found that an irrigation event can reduce the foliage, substrate, and concrete slab temperature by up to 6.4, 4.8, and 1.3 °C, respectively. In this way, the authors demonstrated that the VWC plays a particularly significant role as a temperature regulator in GR systems. In fact, because the VWC affects the thermal mass of the GR, this helps to reduce temperature fluctuations in the system [27]. However, regarding its ecological impact, higher VWC does not always result in better vegetation growth, as observed in humid subtropical climates, where a substrate with lower VWC showed better vegetation cover [28]. Furthermore, concerning thermal behavior, Borràs et al. determined the potential energy savings by installing an irrigation system in a city with a Mediterranean climate (Csa). The authors found that, for this type of climate, the increase in available water does not necessarily imply an increase in total energy savings, i.e., additional irrigation can produce adverse effects on heat dissipation in the GR. Therefore, it is essential to find a balance between reducing water consumption and reducing energy consumption [29]. Wollschläger et al. found that evapotranspiration heat losses in extensive GR without supplemental irrigation are reduced during drought periods. Additionally, the authors identified that a VWC of 9.4% for extensive green roofs generates a balance between cooling efficiency and rainwater absorption (runoff reduction) [30].

This review shows that VWC is a parameter that has a crucial effect on evapotranspiration, as it is one of the primary heat dissipation mechanisms in green roofs [10,31]. Liu et al. mentioned that the moisture behavior of the substrate of GRs located in various climatic regions has not been well understood. Thus, the authors state that further research is needed to fully understand the key factors influencing substrate moisture dynamics and how it affects the thermal performance of GR [32].

Particularly in Mexico, there are studies on implementing GR to reduce energy consumption in buildings. These focus on tropical savannah climate conditions (Aw) [26,33–35], in cold semi-arid climate (BSk) [25], and in study reported by our research group, climates such as hot semi-arid (BSh), hot desert (BWh) and temperate (Cwb) were considered [36]. One of the relevant results in this last work was that the behavior of the volumetric water content (VWC) in the substrate was shown. However, this was presented only for the desert climate city and was done from a global perspective; detailed information on the VWC as a function of the depth of the substrate was not provided. On the other hand, it is important to comment that according to the National Institute of Statistics and Geography [37], the regions of Mexico where there are more air conditioning systems units are the north (48%) and the southeast (12%). This is because, in the northern region, there are climates with extreme dry heat (BSh, BWh), while in the southeast, there are humid and sub-humid tropical climates (Af, Am, Aw). In this way, it is necessary to know the behavior of the GR system under these climate conditions since they are the ones that represent the highest energy consumption for Mexico.

Consequently, while there are studies on the thermal performance of GR in diverse types of climates, few analyze the effect of VWC on energy consumption, and, to the authors' knowledge, there are no studies that report in detail the behavior of VWC in the substrate. This parameter is important because, under specific conditions, VWC can enhance the thermal behavior of GR systems since it can increase evapotranspiration and, therefore, heat loss. Also, it is interesting to investigate the effect of ambient air humidity on evapotranspiration loss, especially in arid climates (low humidity) and in tropical climates (very humid). In fact, in a review article reported by Jamei et al., it is recommended that future research on GR should include more climatic zones and geographical areas with an emphasis on the winter season [38]. Moreover, Gößner et al. acknowledge the complexity of green roof systems and highlight the need for further research to understand evapotranspiration dynamics in different climates and seasons [39].

Due to the above, this work aims to evaluate the GR system in the BWh and Af climates, which can be considered the most representative warm climates of Mexico. Furthermore, these two climates exhibit significantly different characteristics regarding air moisture content. Understanding the impact of ambient air humidity on the thermal behavior of a GR is crucial. This knowledge will enhance our comprehension of how this technology performs, and the insights gained may assist in making informed decisions about its implementation in similar climates.

2. Physical and mathematical model

2.1. Physical model of the green roof

Fig. 1 shows the physical model of the green roof system. It consists of a vegetative layer over a substrate ($H_g = 6$ cm) and a structural roof formed by a concrete slab ($H_{slab} = 10$ cm). Part of the solar radiation passes through the vegetation and reaches the substrate (ground). In this way, much of the solar radiation is absorbed by the ground. Some of this energy is transferred by convection and evaporation to the ambient air, another is transmitted by conduction to the slab, and the rest is stored in the substrate. A part of the energy transferred to the slab is stored on it, and another is transferred to the interior air of an enclosure.

The assumptions taken for the mathematical model are as follows:

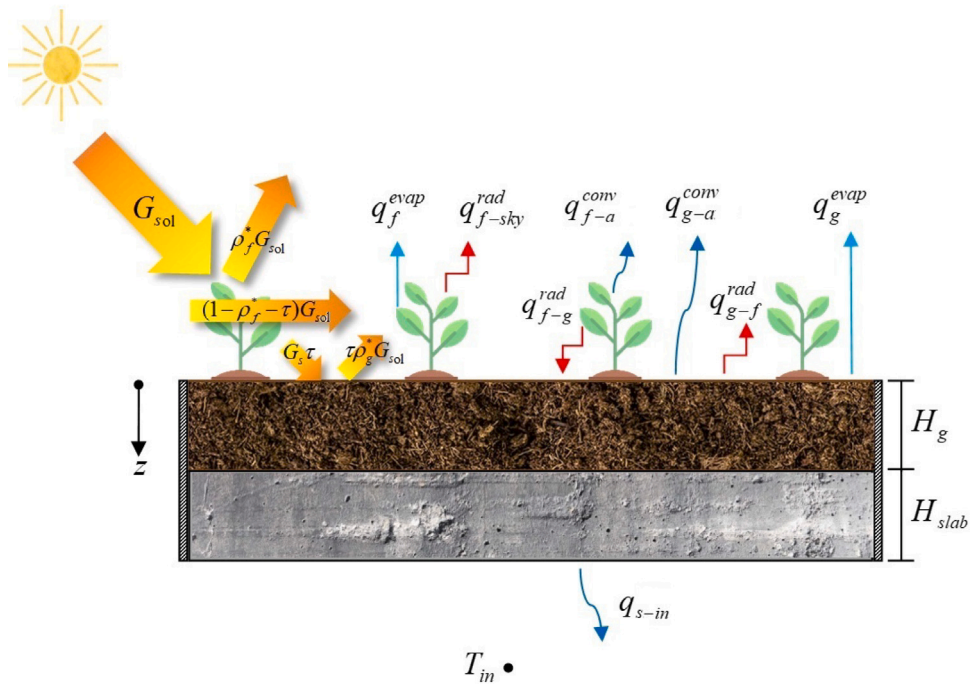


Fig. 1. Green Roof system (GR).

- The GR is big enough to assume horizontal uniformity and apply a one-dimensional analysis (vertical axis).
- The heat and mass transfer in the substrate is considered with a differential and transient state approach.
- In the case of vegetation, the global energy balance approach is considered.
- The use of *aeonium subplanum* plants of the genus *Sedum* of the *Crassulaceae* family is considered.
- The plants and the substrate of the GR have homogeneous properties on the horizontal axis.
- Plants are well irrigated, healthy, and in the full growth stage.
- Plants fully cover the GR substrate.
- The room is kept at a temperature of 25 °C.
- The substrate is a mixture of pumice, compost, and sand, with similar proportions reported by Sailor and Hagos [40]: 50% pumice, 10% compost, and 40% sand.

2.2. Theoretical model of the GR

The solution of the theoretical model of the GR is solved in two sections. The first corresponds to the foliage region that is composed by the plants and the air between them. They are considered healthy, fully grown, pest-free and well irrigated.

2.2.1. Heat transfer mathematical model in the foliage: $T_f(t)$

Of the total solar radiation that reaches in the vegetation (G_{sol}), a fraction is transmitted through it and impinges the substrate ($\tau_f^* G_{sol}$), another fraction is reflected ($\rho_f^* G_{sol}$) and the rest is absorbed by the vegetation ($\alpha_f^* G_{sol}$). The net flux in the foliage due to solar radiation, as well as radiative exchange between the vegetation and surroundings ($R_{f,net}$) is obtained with the following equation:

$$(\rho c_p d)_f \frac{dT_f}{dt} = R_{f,net} - h_{f-a}^{conv} (T_f - T_a) - h_{f-a}^{eva} (P_{f,sat} - P_a) \quad (1)$$

where:

$$R_{f,net} = \alpha_f^* G_{sol} - \epsilon_f^* \sigma (T_f^4 - T_{sky}^4) - \epsilon_{f-g}^* \sigma (T_f^4 - T_g^4) \quad (2)$$

$$\epsilon_{f-g}^* = \left(\frac{1}{\epsilon_f^*} + \frac{1}{\epsilon_g^*} - 1 \right)^{-1} \quad (3)$$

2.2.2. Heat and mass transfer mathematical model in the substrate: $T_g(z, t)$, $\theta(z, t)$

Heat transfer in the substrate is governed by Eq. (4):

$$(\rho c_p)_{g,\theta} \frac{\partial T_g}{\partial t} = \frac{\partial}{\partial z} \left(\lambda_{g,\theta} \frac{\partial T_g}{\partial z} \right) \quad (4)$$

Boundary conditions of Eq. (4) are as follows:

At $z = 0$:

$$-\lambda_{g,\theta} \frac{\partial T_g}{\partial z} \Big|_{z=0} = R_{g,net} - h_{g-a}^{conv} (T_g - T_a) - h_{g-a}^{eva} (P_{g,sat} - P_a) \quad (5)$$

where:

$$R_{g,net} = (1 - \rho_g^*) \tau_f^* G_{sol} + \epsilon_{f-g}^* \sigma (T_f^4 - T_g^4) \quad (6)$$

at $z = H_g$:

$$\lambda_{g,\theta} \frac{\partial T_g}{\partial z} \Big|_{z=H_g} = \lambda_s \frac{\partial T_s}{\partial z} \Big|_{z=H_g} \quad (7)$$

On the other hand, the VWC mathematical model (θ):

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(D_\theta \frac{\partial \theta}{\partial z} \right) + \frac{\partial k_\theta}{\partial z} \quad (8)$$

With the following boundary conditions: at $z = 0$:

$$\left(D_\theta \frac{\partial \theta}{\partial z} + k_\theta \right) \Big|_{z=0} = j_{pr} - j_{eva} \quad (9)$$

where:

$$j_{eva} = \frac{1}{l_{fg}} \left[h_{f-a}^{eva} (P_{f,sat} - P_a) + h_{g-a}^{eva} (P_{g,sat} - P_a) \right] \quad (10)$$

at $z = H_g$:

$$\left(D_\theta \frac{\partial \theta}{\partial z} + k_\theta \right) \Big|_{z=H_g} = \begin{cases} 0, & \theta < \theta_{sat} \\ j_{pr} = j_{dr}, & \theta = \theta_{sat} \end{cases} \quad (11)$$

2.2.3. Heat transfer in the slab: $T_s(z, t)$

Within the concrete slab, there is only conductive heat transfer, and this process is described by the Eqs. (12)–(14):

$$(\rho c_p)_s \frac{\partial T_s}{\partial t} = \frac{\partial}{\partial z} \left(\lambda_s \frac{\partial T_s}{\partial z} \right) \quad (12)$$

$$\lambda_s \frac{\partial T_s}{\partial z} \Big|_{z=H_g} = \lambda_{g,\theta} \frac{\partial T_g}{\partial z} \Big|_{z=H_g} \quad (13)$$

$$-\lambda_s \frac{\partial T_s}{\partial z} \Big|_{z=H_s} = h_{in} (T_s - T_{in}) \quad (14)$$

2.3. Heat transfer in the RR slab: $T_r(z, t)$

The RR is a concrete slab with a thickness of 10 cm. Similar to the GR, in the RR, the heat flow is considered to be occur only in the z -direction. The outer surface of the RR is exposed to solar radiation (G_{sol}) and presents convective and radiative losses. On the other hand, the inner surface of the RR presents convective exchange with the building. Thus, the heat transfer mathematical model for the RR is given by the Eqs. (15)–(17):

$$(\rho c_p)_r \frac{\partial T_r}{\partial t} = \frac{\partial}{\partial z} \left(\lambda_r \frac{\partial T_r}{\partial z} \right) \quad (15)$$

$$-\lambda_r \frac{\partial T_r}{\partial z} \Big|_{z=0} = \alpha_r^* G_{sol} + h_{r-a} (T_r - T_a) + \epsilon_r^* (T_r^4 - T_{sky}^4) \quad (16)$$

$$-\lambda_r \frac{\partial T_r}{\partial z} \Big|_{z=H_r} = h_{in} (T_r - T_{in}) \quad (17)$$

Table 2 shows the thermophysical and radiative properties of the system, as well as some other parameters. The convection and diffusion correlations are detailed in Appendices A and B. Finally, it is important to note that, in all cases, the initial conditions of the theoretical model correspond to the environmental conditions at 00:00 h (Figs. 4 and 5). Also, the substrate is considered to be initially saturated.

Table 2
Properties and parameters used in the simulation.

System	Component	ϵ^*	ρ^*	τ^*	λ (W/mK)	ρ (kg/m ³)	c_p (J/kgK)	Thickness (m)	LAI
GR	Concrete	0.40	0.82	- - -	1.74	2300	840	0.10	- - -
	Substrate	0.15	0.95	- - -	0.25	1370	800	0.06	- - -
	Water	- - -	- - -	- - -	0.60	- - -	4180	- - -	- -
	Foliage	0.28	0.90	0.10	- - -	- - -	- - -	0.08	0.50
RR	Concrete	0.40	0.82	- - -	1.74	2300	840	0.10	- - -

2.4. Study parameters

The thermal parameters analyzed in this study include the temperatures of various components within the systems and the thermal energy gained through the roof (q_{s-in}). The value of q_{s-in} is calculated using Eqs. (14) and (17) for the GR and RR, respectively. Additionally, Eq. (18) is used to determine the net heat gain over the evaluation period.

$$Q_{gain} = \int_{\text{day } 1}^{\text{day } 3} q_{s-in}(t) dt \quad (18)$$

Finally, another important parameter to consider is the VWC. This parameter is crucial for the thermal behavior of the GR. It modifies the substrate's thermophysical properties, influencing heat and mass transfer. For a more detailed analysis, three positions of the VWC in the substrate are considered: at the bottom, at a midpoint, and on the outer surface. By analyzing the thermal parameters alongside the VWC, we can gain insights into the overall impact on heat transfer within the system.

3. Solution methodology

3.1. Numerical solution

The governing equations were solved numerically with a code developed in the Fortran language. On the one hand, the heat transfer mathematical model in the foliage (Eq. (1)), by representing a global energy balance, only requires the transient term discretization using an implicit scheme. Secondly, the discretization process of the heat and mass transfer governing equations in the substrate, concrete slab, and reference roof involves discretizing the general diffusion equation with the finite volume method. This process is described by Xamán and Gijón [41]. The resulting equation system in the discretization process is solved with the tridiagonal matrix algorithm (TDMA) employing a Gauss–Seidel approach. Additionally, to give more stability to the algorithm, the radiative terms were linearized according to Patankar's recommendation [42].

The GR system is resolved in three blocks. The first corresponds to the heat transfer in the foliage, followed by the heat and mass transfer in the substrate, and for the third block, the heat transfer in the concrete slab is solved. The three blocks are coupled sequentially as illustrated in Fig. 2. Initially, the known parameters are entered, the code reads the climatic data and subsequently, the computational mesh is built (step 1). In the next step, time is increased (step 2), then the foliage section is solved (step 3), followed by the solution of the substrate section (step 4) and finally, the conductive model of the slab (step 5). The procedure is repeated from step 2 to 5 until it reaches the convergence criteria, which was set as 10^{-5} . After achieving convergence, the solution is stored (step 6). Then, the procedure is repeated from step 2 to 6 until reaching the final simulation time.

To verify the precision of the numerical results, the climatic data of November 21, 2016, reported by Chagolla-Aranda et al. are considered [26]. It was determined that $\Delta t=1$ s and $Nz=107$ produce time-step and mesh density-independent results for the GR. In the RR case, the appropriate step-time and node value was $\Delta t=1$ s and $Nz=67$.

3.2. Theoretical model validation of the GR and RR

In order to validate the numerical code, the experimental study results of the heat and mass transfer in a GR system reported by Chagolla-Aranda et al. were reproduced [26]. Fig. 3 compares numerical and experimental data for the surface substrate temperature (Fig. 3a) and VWC (Fig. 3b) throughout three days. For the substrate temperature, it was found that an average difference between numerical and experimental of 0.46 °C and that 89.1% of the calculated VWC were significantly close to the experimental measurements with a $\pm 5\%$ during the entire analyzed period. On the other hand, in the VWC comparison, for a 3 cm depth, an average difference of 0.007 m³/m³ was obtained, and 98% of the calculated temperatures are significantly close to the experimental measurements with $\pm 5\%$ during the three simulated days.

Finally, a root mean square deviation of 0.6 was found for the substrate temperature and 0.17 for the VWC. These values indicate the model's strong predictive capabilities and its effectiveness in simulating real-world conditions. Based on these results, it is determined that the theoretical model can effectively be utilized to study the thermal behavior of a green roof (GR) system under real conditions, as it accurately reproduces the experimental evidence.

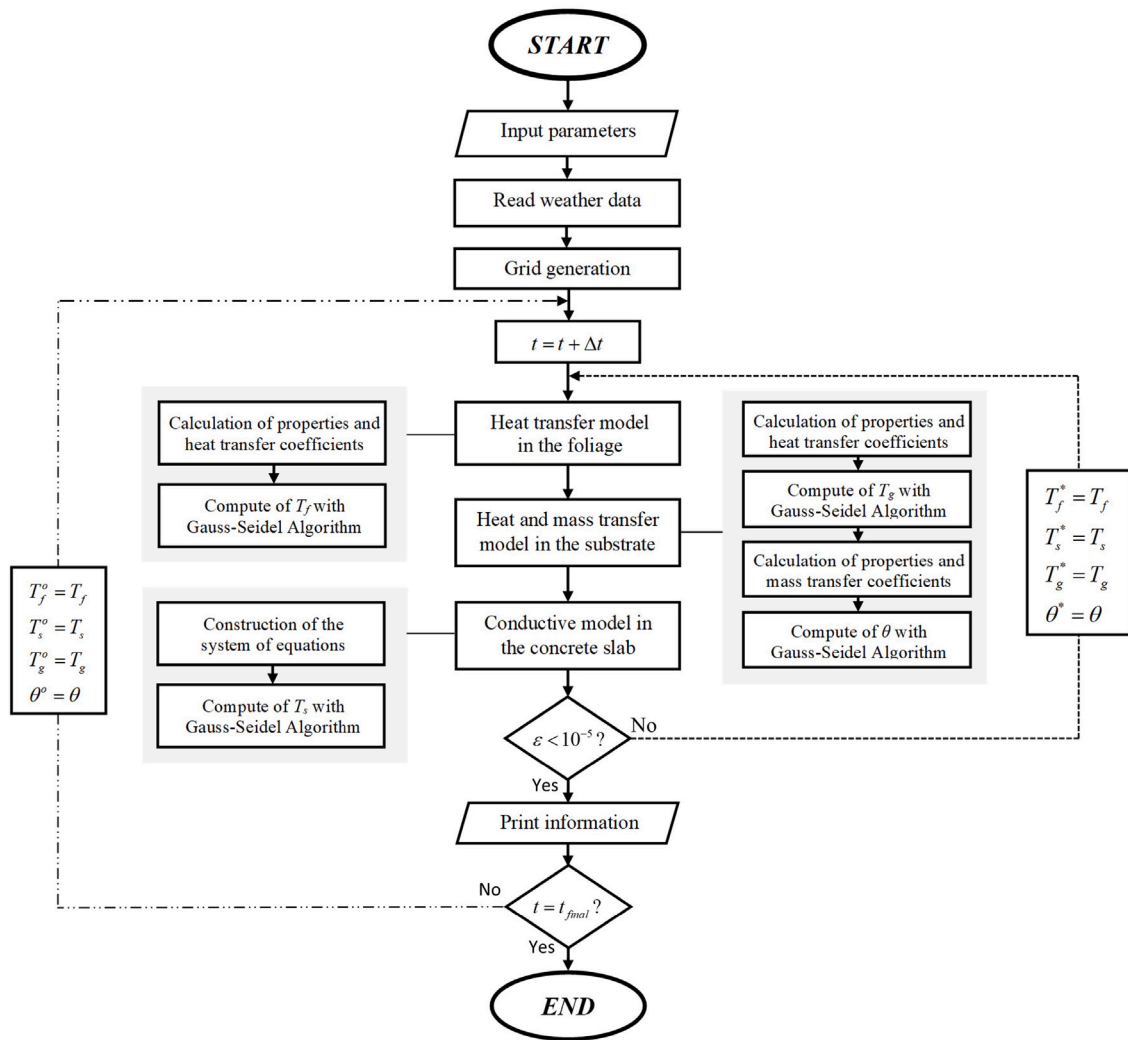


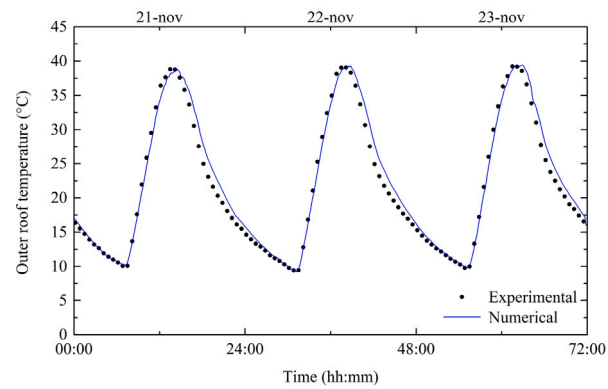
Fig. 2. Flowchart of the numerical procedure.

3.3. Climate data of the considered cities

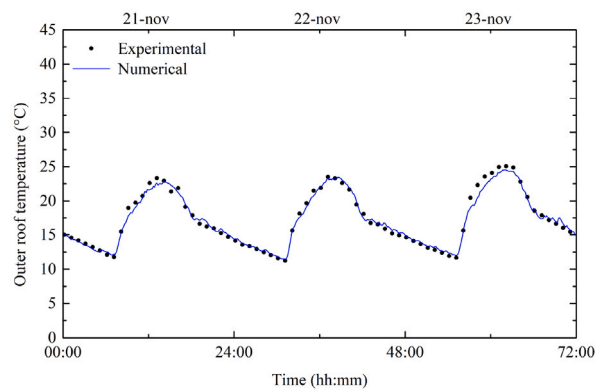
For representing a warm humid climate, the city of Villahermosa, Tabasco, located in the southwestern region of Mexico, was selected. Situated at an elevation of 9 masl, it has an average annual temperature of 27 °C, and receives an average annual rainfall 2550 mm, with precipitation occurring throughout the year, especially during the summer months. The Köppen classification for this city limits between Af and Am . Fig. 4 illustrates for both the warm and cold periods. During the warm period (April 27–29) the maximum temperature reached 39 °C, accompanied by peak solar radiation of 1035 W/m². Humidity and wind speed recorded values of 67% y 1.4 m/s, respectively. In contrast, the cold period (January 29–31) exhibited stable temperature ranges between 16 and 28 °C, with a maximum solar radiation of 607 W/m². Average relative humidity and wind speed during this period were 91% y 2.1 m/s, respectively.

On the other hand, Hermosillo, Sonora, was selected as a representative city with a desert climate. During the warmest period of the year (June 2–4), the maximum temperature reaches 49, while the minimum temperature is 23 °C. The maximum solar radiation during this time is 1031 W/m², with a relative humidity of 18% and an average wind speed of 2.5 m/s. In contrast, the climatic conditions during the coldest period, (December 26–28) show a maximum temperature of 23.1 °C, and a minimum temperature of 2.1 °C. The maximum solar radiation recorded during this period is 741 W/m², with a relative humidity of 38% and an average wind speed of 1.8 m/s. The evolution of these variables are illustrated in Fig. 5.

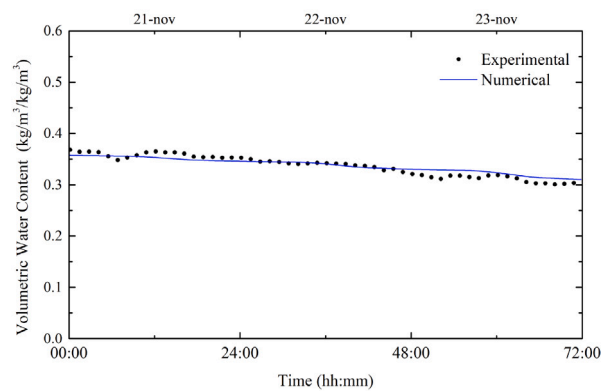
The climatic data of both cities were taken from the databases of the meteorological stations that belong to the National Meteorological Service and the National Water Commission (SMN-CONAGUA) [43].



(a) Outer temperature (RR)



(b) Outer temperature (GR)



(c) Volumetric water content (GR)

Fig. 3. Theoretical model validation with Chagolla et al. experimental results [26].

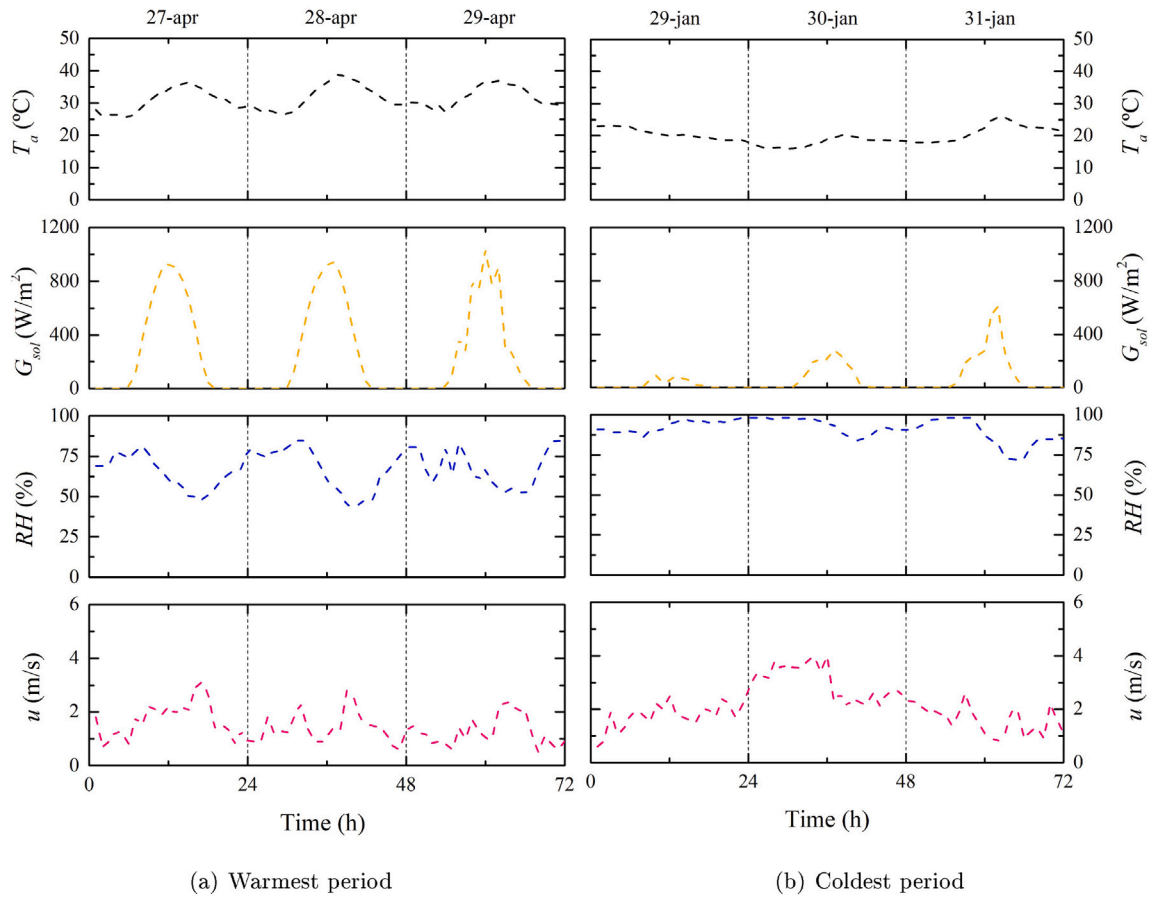


Fig. 4. Climate data for Villahermosa.

4. Results and discussion

4.1. Results for tropical climate

4.1.1. Warmest period (Af/Am)

Fig. 6 illustrates the thermal response of both systems during the warm period. For the GR system, the data includes ambient temperature (T_a), foliage temperature (T_f), surface substrate temperature (T_{g-top}), inner surface of the slab temperature (T_{s-in}) and the VWC. In contrast, for the RR system, the information presented are the exterior surface temperature (T_{s-ext}) and the interior slab temperature (T_{s-in}). Additionally, the heat gain in both systems is presented.

Figs. 6a and 6b allow to appreciate the benefit of using a GR system. T_{s-in} of the GR system presents a moderate behavior compared to the RR system since it stays between 27 and 34 °C, unlike RR, whose interior slab temperature presents values between 26 and 54 °C. The foliage temperature (T_f) tends to the value of T_a all over the period, and the temperature of the exterior surface of the substrate (T_{g-top}) keeps below T_a . Fig. 6c displays the heat gain to the building in both systems, where positive values indicate energy gains and negative values indicate energy losses to the environment. The RR system only has energy gain, reaching a maximum of 150 W/m² on day 2. On the other hand, the GR system experiences both gains and losses. The interior heat gain is moderate, reaching a maximum of 19 W/m², while heat losses can be as high as 43 W/m². Over the evaluation period, the net heat gain (Q_{gain}) for the GR system is −227.8 J/m², whereas the RR system shows a substantial gain of 3300.39 J/m². This is a direct result of the fact that the RR system is consistently lossless. Finally, Fig. 6d shows the VWC evolution at three points in the substrate: the top surface (θ_{top}), the middle (θ_{mid}) and the bottom (θ_{bot}). It is evident that there are no significant differences in VWC among these three positions. A downward linear trend is noted throughout the observation period, with VWC decreasing from 0.44 to 0.36. This represents an 18% reduction, which can be considered moderate.

4.1.2. Coldest period (Af/Am)

Figs. 7a and 7b allow to compare the thermal behavior of the GR and RR systems. The GR system exhibits a moderate response in T_f , T_{g-top} and T_{s-in} . During the coldest period, T_f tends to match T_a , just as it does during the warmest period. However, T_{g-top}

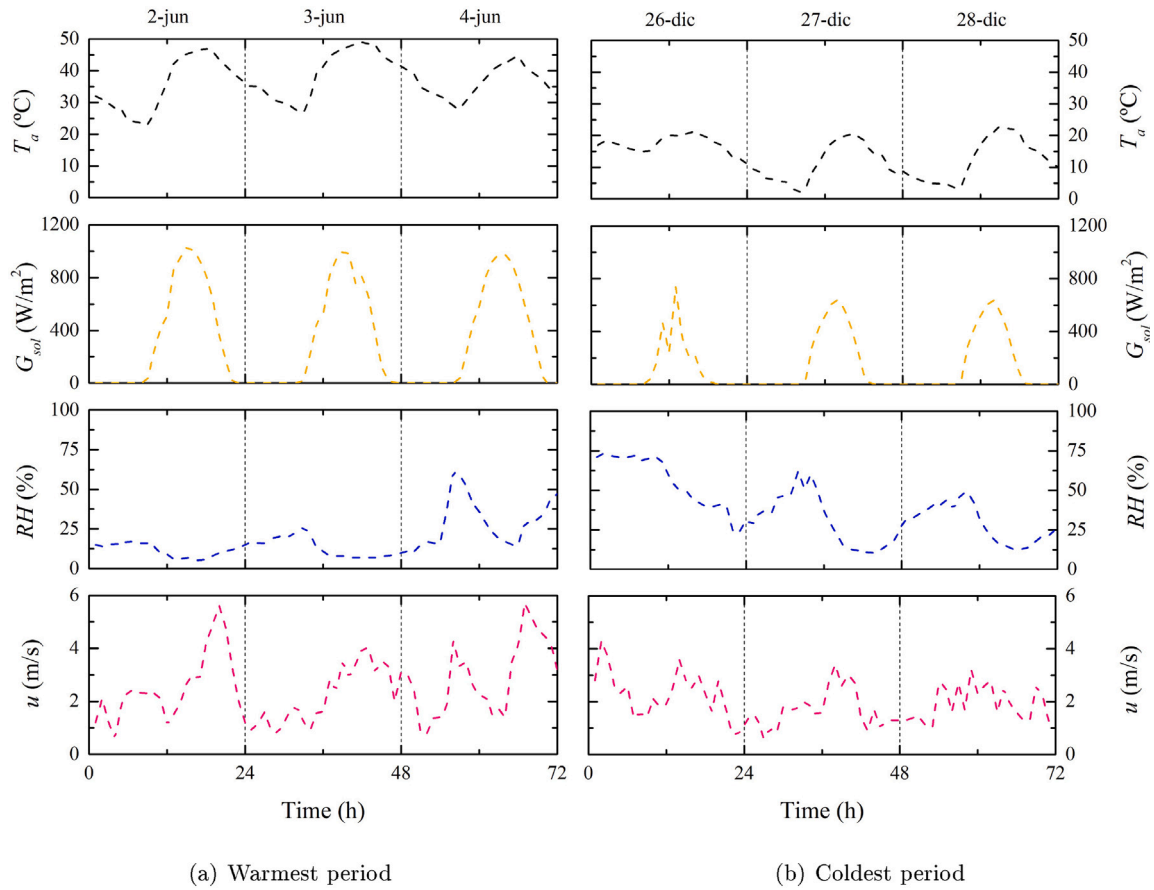


Fig. 5. Climate data for Hermosillo.

and T_{s-in} , which usually remain close to the ambient temperature, show a difference between 8:00 AM to 8:00 PM. During this time frame, both T_{g-top} and T_{s-in} are maintained up to 3 and 5 °C below T_a , respectively. Instead, the slab temperatures in the RR system are typically slightly higher than T_a , especially in the afternoons. On day 3, the maximum temperatures were 37 °C for T_{s-ext} and 33 °C for T_{s-in} . Fig. 7c shows that all day 1 and until 9:00 AM of day 2, the heat gain is moderate and very similar to both systems. However, during the rest of the cold period, the RR system reaches higher values than the GR system. In fact, the GR system experiences only losses to the exterior. As a result, the net heat gain to the building (Q_{gain}) for the GR system is 59.12 J/m², while for the RR system, it is 908.89 J/m². Finally, Fig. 7d shows that the VWC in the substrate presents insignificant variations during the cold period, with a reduction of 0.45 to 0.43 (4.4%). A comparative analysis with the relative humidity in the air, as illustrated in Fig. 4b, suggests that this behavior occurs because the air is nearly saturated with humidity, which inhibits moisture loss from the substrate. Therefore, this suggests that the energy losses of the GR system are primarily due to external convective and radiative components.

4.2. Results for hot desert climate

4.2.1. Warmest period (BWh)

Fig. 8 presents the behavior of several parameters in the GR and RR systems. For the GR system, the foliage temperature (T_f), the external surface of the substrate (T_{g-top}), and the slab interior surface are analyzed. Additionally, for the RR system, the slab exterior and interior temperatures are considered (T_{s-ext} , T_{s-in}). In Figs. 8a and 8b, the thermal behavior of the GR system is moderate. Throughout the three days, T_f and T_a have no significant differences. On the other hand, the interior slab temperature (T_{s-in}) presents maximum values of 42.14 (day 1), 45.83 (day 2) and 40.36 °C (day 3). Further, the substrate surface temperature (T_{g-top}) is always below T_a . Initially, there is a difference of 14 °C (day 1). However, this difference diminishes with time until the beginning of day 3, where T_{g-top} reaches values close to T_a . Instead, the RR system has higher values than GR. T_{s-ext} reaches a maximum value of 69.18 (day 1), 69.04 (day 2) and 66.3 °C (day 3), whilst T_{s-in} presents a maximum value of 59.22 (day 1), 60.16 (day 2), and 56.62 °C (day 3). For the hot season of this climate, it is intended that the temperature in the lower part of the roof (T_{s-in}) be as low as possible both during the day and at night. A comparison of T_{s-in} between GR and RR systems shows that the GR system

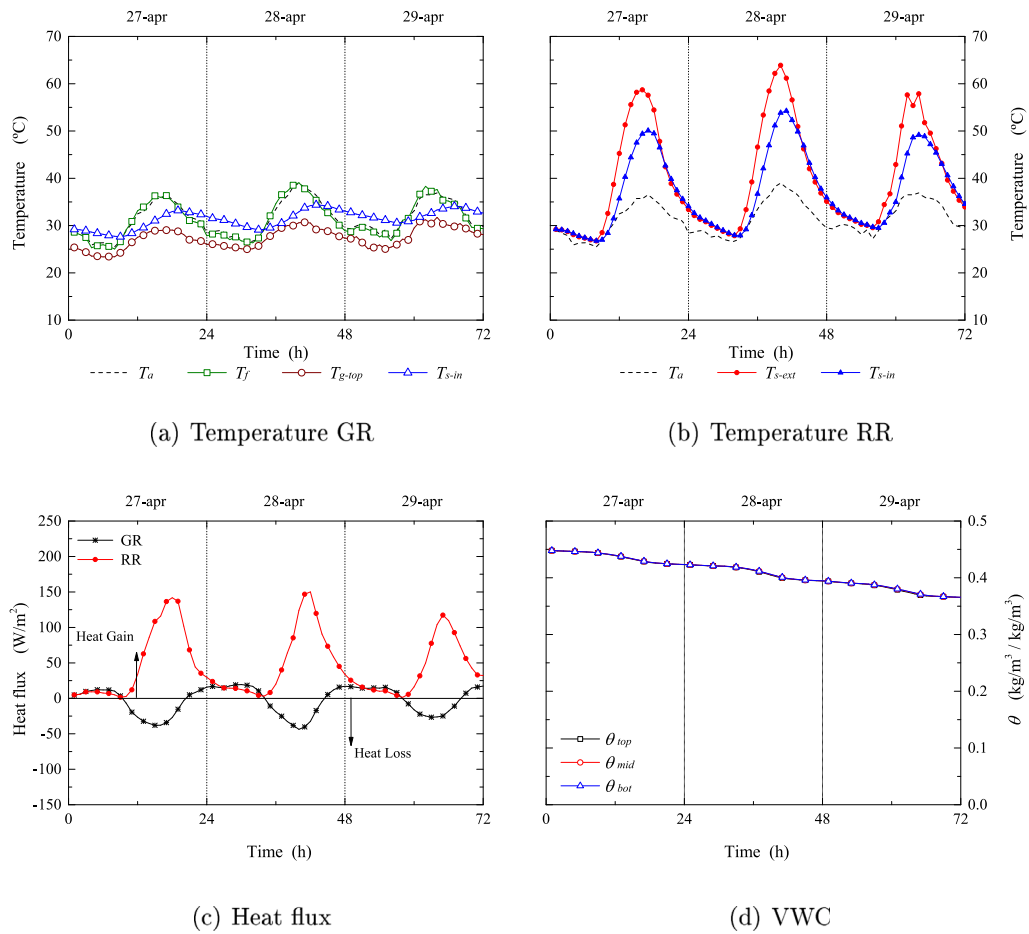


Fig. 6. Behavior of thermal parameters for warmest period (Tropical).

better meets this requirement than the RR system. Fig. 8c shows the heat flow into the building interior for both systems. In the case of the RR system, energy losses are minimal during the first two days, with no losses recorded on the third day. This indicates that the RR system only experiences energy gains. During the three simulation days, there is a significant increase in heat gain that begins at noon, reaching its peak at different times each day. On Day 1, the maximum heat gain occurs at 7:00 PM (200.31 W/m²). On Day 2, it peaks at 5:00 PM (159.22 W/m²), while on Day 3, it reaches the maximum value at 7:00 PM (161.20 W/m²). Once the Sun goes down, the rate of heat gain decreases rapidly. In contrast, after midnight, this decrease becomes much more gradual. On the other hand, the heat gain in the GR system was generally lower than the heat losses, particularly on Day 1. On this day, the maximum gain is 58.58 W/m² at 7:00 AM, while the maximum loss was 99.85 W/m² at 3:00 PM. For Day 2, the maximum gain and loss values were 49.76 W/m² at 7:00 AM and 79.30 W/m² at 2:00 PM, respectively. On Day 3, the maximum heat gain reached 80.99 W/m² at 8:00 AM, and the maximum loss was 55.86 W/m² at 4:00 PM.

It is important to note that the peak values of these gains occur at different periods, indicating a time lag between the energy peaks of both systems. In the GR system case, maximum energy gain occurs around 8:00 AM, while for the RR system, it happens near 7:00 PM. The observed behavior can be attributed to the thermal mass of the components in the GR system, which is greater than that of the RR system. This higher thermal mass causes a delay in the propagation of thermal energy throughout the system. Additionally, the GR system provides thermal buffering for q_{s-in} . As a result, the net energy gain to the building (Q_{gain}) in RR system is 4460.53 J/m², while for the GR system is -172.40 J/m². This negative value clearly indicates that the GR system fundamentally operates as an energy dissipator. Similar to the Af climate case, this demonstrates the effectiveness of the GR system in reducing energy gain in buildings.

Finally, Fig. 8d displays the volumetric water content in the top substrate surface (θ_{top}), the bottom (θ_{bot}), and the middle point (θ_{mid}). From Day 1 until the morning of Day 2, VWC is the same in the three points. However, θ_{top} starts to reduce at 10:00 AM of day 2. This decline is most significant during sunny hours: from 10:00 AM to 7:00 PM on Day 2, and from 7:00 AM to 4:00 PM on Day 3. Instead, this reduction is negligible during the nighttime periods between day 2 and day 3. The observed behavior on the upper surface of the substrate is due to the low humidity in the air, which is typical of summer in BWh climates. It is important to note that from 4:00 PM on day 3, the VWC on the outer surface of the substrate begins to increase. This increase is due to moisture

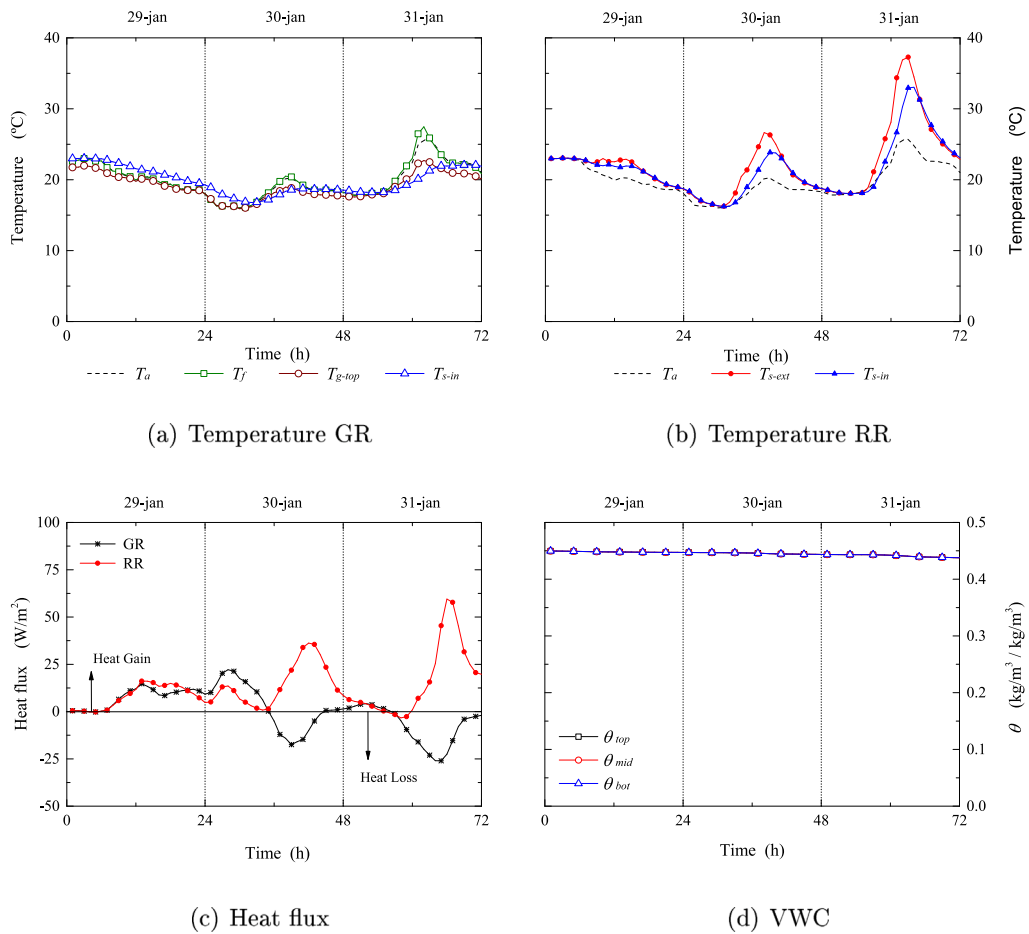


Fig. 7. Behavior of thermal parameters for coldest period (Tropical).

being transported through diffusion from the lower layers to the upper layers of the substrate. In fact, the VWC decreased from 0.45 to 0.17, which represents a 62% reduction; this indicates a significant decline in the VWC. At this point it would be advisable to irrigate the substrate to maintain the continuity of evaporative losses to the environment.

4.2.2. Coldest period (BWh)

Figs. 9a and 9b illustrate the temperatures for both roof systems. Similar to the hot period, T_f remains the same as T_a throughout the cold period. Both temperatures fluctuate between 2.4 and 23 °C. The external surface temperature of the substrate (T_{g-top}) consistently stays below T_a , ranging from a minimum of 2 °C to a maximum of 17 °C. For Day 1, T_{s-in} exhibits moderate behavior, remaining between 17 and 20 °C. However, during days 2 and 3, T_{s-in} significantly drops to a range of 9 and 17.4 °C. In the RR case, for the first nine hours of Day 1, both the exterior surface temperature (T_{s-ext}) and interior surface temperature (T_{s-in}) of the slab remain similar to T_a . However, both T_{s-ext} and T_{s-in} maintain values higher than T_a except between 8:00 AM and 1:00 PM on days 2 and 3. During this time, T_{s-ext} reaches a maximum of 36 °C, while T_{s-in} reaches 31 °C, placing them between 4 and 9 °C above T_a .

Fig. 9c presents the evolution of the heat flow to the building interior for both systems. The heat gain in the RR system is significantly higher than that in the GR system. In contrast to the RR system, the GR system experiences substantial heat dissipation to the outside, reaching up to 64.6 W/m² on Day 3. For the RR system, these losses are almost negligible on Day 1 and moderate on days 2 and 3. Furthermore, similar to the warm period, the RR system achieves its peak heat gain values when the GR system experiences its highest losses. Specifically, the GR system has a heat gain (Q_{gain}) of 58.96 J/m², while the RR reaches 1919.42 J/m². Therefore, under the conditions and considerations raised, the GR is also more suitable for reducing the thermal load. In Fig. 9d, the VWC of the substrate for the GR system is shown. It is observed that the decrease in humidity is minimal; the VWC reduced from 0.45 to 0.37 which is a 17% decline over the period. This reduction is primarily due to the cold period, which has higher relative humidity in the ambient air compared to warmer days. Therefore, as the humidity in the air increases, heat loss from the substrate due to evaporation and the evapotranspiration of foliage decreases.

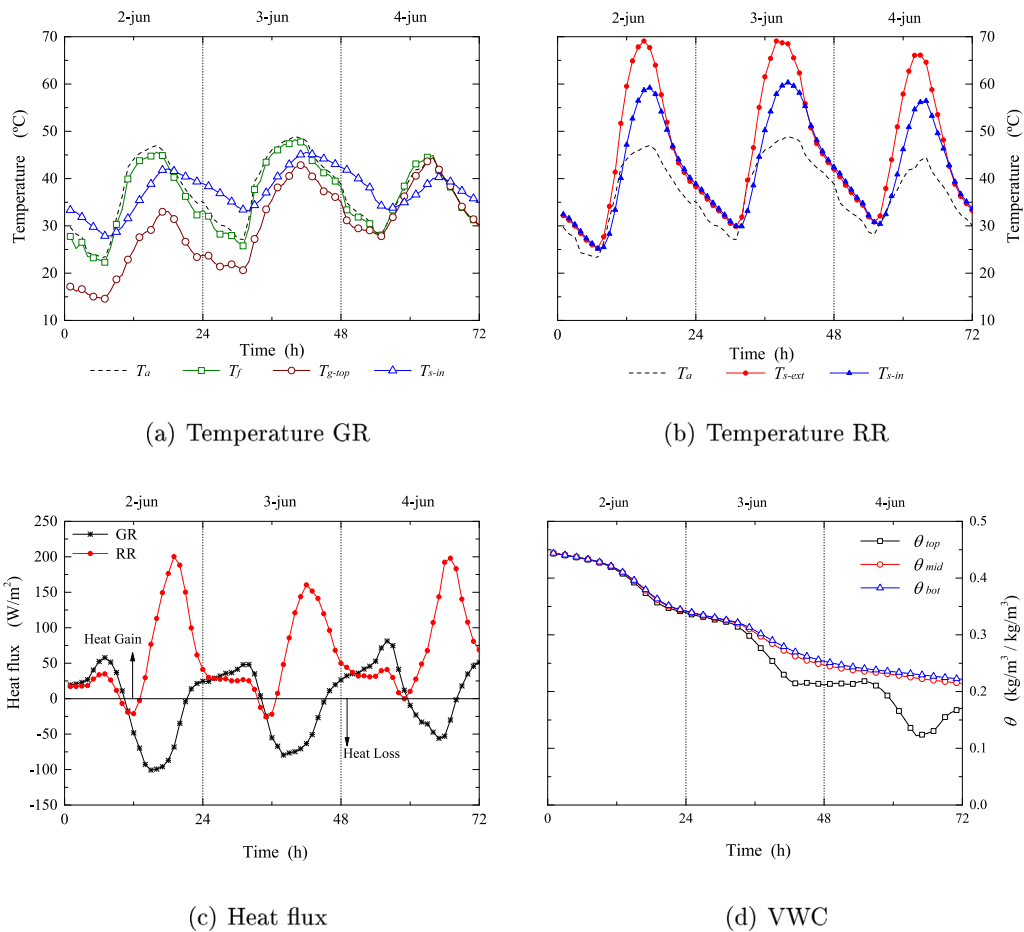


Fig. 8. Behavior of thermal parameters for warmest period (Dry).

5. Conclusions

A model was implemented to solve the heat and mass transfer on a green roof system (GR). This model was then used to numerically predict the performance of the GR system under the climatic conditions of two Mexican cities with warm climates: Hermosillo, Sonora (*BWh*) and Villahermosa, Tabasco (*Af/Am*). With the purpose of evaluating the performance of the GR system, the obtained results were compared with a conventional roof system. To evaluate the performance of the GR system, the results were compared with those of a conventional roof system. From these findings, the following conclusions can be drawn:

- The theoretical model accurately predicts heat and mass transfer in the green roof system, as it closely matches experimental results, especially the VWC in the substrate, achieving a mean square error of just 0.6%.
- During both the coldest and warmest periods, the foliage temperature remains close to the ambient air temperature. Furthermore, this behavior is independent of the climates considered.
- Despite the high humidity commonly found in tropical climates, the GR system consistently reduces energy gains during the hottest periods by up to 94%. Furthermore, during the coldest periods, all components of the GR system effectively maintain a temperature very close to the ambient air, around 22 °C.
- In the tropical climate, the volumetric water content of the substrate showed no dependence on depth. Additionally, the VWC of the substrate was reduced by only 18.1%. During the warm period, the VWC of the substrate decreased by only 18.1%. In contrast, during the cold period, the VWC remained virtually constant, with a reduction of less than 4.5%.
- In both the warmest and coldest periods of the *BWh* climate, the GR system presents predominantly losses. This significantly reduces the net heat gain, with relative differences exceeding 90% when compared to the RR.
- In the *BWh* climate, the volumetric water content in the substrate changes significantly depending on the season. During the coldest period, there is a moderate reduction of 17.1% in VWC, with no noticeable differences across the various layers. However, in the hottest period, the top layer of the substrate tends to dry out more severely, with a reduction of over 60%. It is advisable to irrigate the substrate at noon on the third day to allow energy dissipation by evaporation.

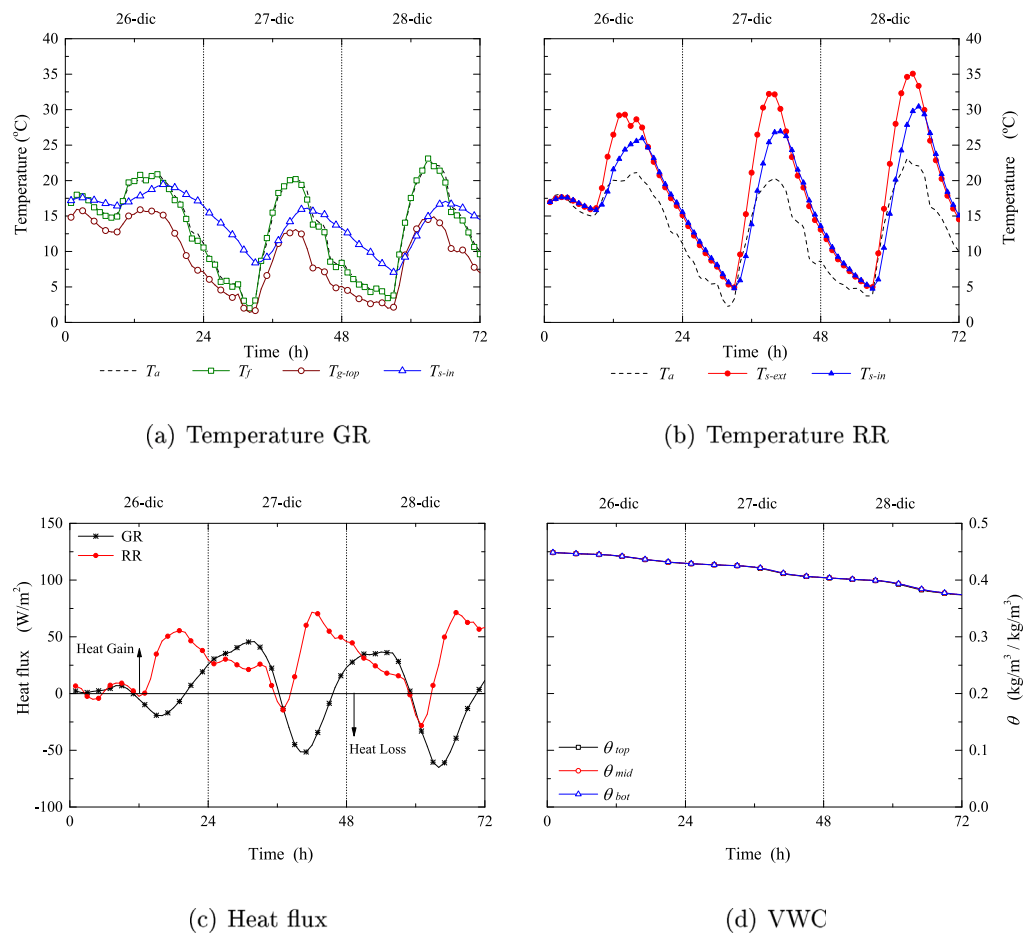


Fig. 9. Behavior of thermal parameters for coldest period (Dry).

It can be concluded that the GR system performs satisfactorily in cities with *BWh* and *Af/Am* climate, significantly reducing energy consumption required to maintain thermal comfort conditions in buildings, especially in hot periods. However, the findings of this study can be applied to a GR system under typical climatic conditions, i.e. non-extreme climatic conditions. It is important to note that the model assumes the substrate is initially saturated and does not address irrigation requirements. Therefore, future research should assess the performance of the GR system under extreme climate scenarios, such as heatwaves and heavy rainfall. Additionally, given the low energy gain from the GR system, it is important to evaluate its performance under free evolution conditions. This will allow determine the impact of the cold period on the performance of a GR, particularly on the *BWh* climate, which is characterized by harsh winters. Finally, it is recommended to conduct research on the impact of irrigation on energy consumption in arid climates, especially when hydraulic pumping is necessary.

CRedit authorship contribution statement

M.A. Chagolla-Aranda: Writing – original draft, Methodology, Investigation, Conceptualization. **E. Simá:** Resources, Funding acquisition. **I. Hernández-López:** Writing – review & editing, Visualization, Software. **A. Piña-Ortiz:** Writing – review & editing. **A. Ávila-Hernández:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Correlations for foliage convection.

$$h_{f-a}^{conv} = LAI \frac{(\rho c_p)_a}{r_a} \quad (A.1)$$

The aerodynamic resistance r_a is obtained from empiric correlation [44]:

$$r_a = \frac{1}{k^2 u} \ln \left(\frac{z - d_0}{Z_0} \right) \quad (A.2)$$

where:

$$d_0 = 0.701 H_f^{0.975} \quad (A.3)$$

$$Z_0 = 0.131 H_f^{0.997} \quad (A.4)$$

$$h_{f-a}^{eva} = \frac{LAI}{\gamma} \frac{(\rho c_p)_a}{r_a + r_{stoma}} \quad (A.5)$$

Djedjig et al. recommend the r_{stoma} determination with the following relation [45]:

$$r_{stoma} = \frac{r_{stoma,min}}{LAI} f_1(G_{sol}) f_2(T_f) f_3(P_f) f_4(\theta) \quad (A.6)$$

where:

$$f_1(G_{sol}) = 1 + \exp[-0.034(G_{sol} - 3.5)] \quad (A.7)$$

$$f_2(T_f) = \frac{\exp[0.3(T_f - 273)] + 258}{\exp[0.3(T_f - 273)] + 27} \quad (A.8)$$

$$f_3(P_f) = 4 \times 10^{-3} + \exp \left[-0.73 \frac{0.622 \times 10^3}{P_{atm}} (P_{f,sat} - P_a) \right] \quad (A.9)$$

$$f_4(\theta) = \frac{\theta_{sat} - \theta_{min}}{\theta - \theta_{min}} \approx \frac{\theta_{sat}}{\theta} \quad (A.10)$$

Appendix B

Correlations for convection and diffusion in the substrate.

$$h_{g-a}^{conv} = LAI \frac{(\rho c_p)_a}{r_a + r_c} \quad (B.1)$$

$$h_{g-a}^{eva} = \frac{LAI}{\gamma} \frac{(\rho c_p)_a}{r_a + r_{vap}} \quad (B.2)$$

The aerodynamic resistance r_c can be obtained from the empiric correlation proposed by Campbell and Norman [46].

$$r_c = \frac{1}{0.004 + 0.012 u_g} \quad (B.3)$$

where the velocity over the substrate surface is obtained with [47]:

$$u_g = u_f \exp \left[-a \left(1 - \frac{0.05}{H_f} \right) \right] \quad (B.4)$$

$$u_f = u \left[\ln \left(\frac{H_f - d_0}{Z_M} \right) / \ln \left(\frac{Z_u - d_0}{Z_M} \right) \right] \quad (B.5)$$

$$a = (0.28 LAI H_f d_f)^{0.5} \quad (B.6)$$

r_{vap} is obtained with the following correlation [48]:

$$r_{vap} = c_0 + c_1 \left(\frac{\theta}{\theta_{sat}} \right)^{-c_2} \quad (B.7)$$

The c_0 , c_1 and c_2 coefficients can be found in the Camillo and Gurney work [49].

$$(\rho c_p)_{\theta} = c_{p,\theta} (0.2 + \theta) \rho_g \quad (B.8)$$

$$\lambda_{g,\theta} = \lambda_{\theta,dry} + \theta k_{\theta} \quad (B.9)$$

$$k_{\theta} = k_{\theta,sat} \left(\frac{\theta}{\theta_{sat}} \right)^{2b+3} \quad (B.10)$$

$$D_{\theta} = - \frac{b k_{\theta,sat} \psi_{sat}}{\theta} \left(\frac{\theta}{\theta_{sat}} \right)^{b+3} \quad (B.11)$$

The $k_{\theta,sat}$, ψ_{sat} , θ_{sat} and b values for several substrate types are reported by the agriculture department of the United States of America [50]. The values considered in this work are $k_{\theta,sat}=7.0E-06$ m/s, $\psi_{sat}=0.48$ m, $\theta_{sat}=0.45$ kg/m³/kg/m³ and $b=5.39$.

Data availability

Data will be made available on request.

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